

## **Effectiveness of Teaching Approaches on Foundational Learning in Emergency Contexts in Sub-Saharan Africa: A Systematic Review Protocol**

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**Registration:** This protocol will be registered on PROSPERO, the International Prospective Register of Systematic Reviews, prior to commencing data collection. The registration number will be inserted in the final published review. The protocol will also be submitted for title registration with the Campbell Collaboration Education Coordinating Group.

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**Ethical approval:** Ethical approval is not required for this study. The review involves secondary analysis of published and publicly available literature only. No primary data will be collected from human participants.

**Conflict of interest:** The authors declare no conflicts of interest. Members of the review team have been involved in establishing the Communities of Practice and the Regional Learning Agenda that identified this research question as a priority, but have not been involved in the design, delivery, or evaluation of any specific teaching approach or professional development intervention under review.

**Reviewer roles:** Content expertise is provided by Moses Awazi Bah, Carine Mbah, Clifort Jong Yunji, Vivian Fonkwo Mekwatche, and Gam Linda Ankiambom. Methodological leadership and guidance on JBI systematic review methodology is provided by Mark Tata Kelse, Claude Dzelamonyuy Fotso, and Rigobert Pambe. Information retrieval and search strategy design will be led by Sherifa-Jamil Nashipu. Statistical analysis and meta-analytic synthesis will be a shared task under methodological oversight. Any unresolved disagreements during screening or data

extraction will be referred to Patrick Okwen (eBASE Africa), who will serve as independent adjudicator.

## **Abstract**

### ***Background***

Foundational literacy and numeracy are essential for all subsequent learning, yet across Sub-Saharan Africa, millions of children are learning, or failing to learn, in classrooms disrupted by armed conflict, forced displacement, and humanitarian crisis. The region hosts over half of the world's crisis-affected children, and existing evidence suggests that children in crisis-affected settings may acquire foundational skills up to six times more slowly than peers in stable contexts (UNICEF Eastern and Southern Africa, 2023). Despite the scale of this challenge, the evidence base on which teaching approaches actually improve foundational learning outcomes in emergency settings is fragmented, unsynthesised, and largely inaccessible to the practitioners and policymakers who need it most.

### ***Objective***

### ***Inclusion criteria***

contextual factors.

### ***Methods***

### ***PROSPERO registration number***

## Introduction

Foundational literacy and numeracy encompass the basic reading, writing, and numeracy competencies that every child should acquire in the early years of schooling. They are widely recognised as the prerequisite infrastructure through which all subsequent learning occurs, and the bedrock of a child's long-term educational and economic prospects (UNICEF, 2022; World Bank, 2023). Despite decades of investment in school enrolment across Sub-Saharan Africa, a profound learning crisis persists. The World Bank (2023) estimates that nearly six in ten children in low- and middle-income countries experience learning poverty, defined as the inability to read and understand a simple text by the age of ten. Sub-Saharan Africa sits at the epicentre of this crisis: nine out of ten ten-year-olds in the region cannot read with basic comprehension, the highest rate of learning poverty globally, while approximately 84% are not proficient in foundational mathematics (ADEA, 2024).

This baseline crisis is dramatically compounded by the frequency and severity of humanitarian emergencies across the region. Sub-Saharan Africa hosts over half of the world's crisis-affected school-age children, exposed to armed conflicts, forced displacement, climate-induced disasters, and health epidemics that destabilise already fragile education systems (UNICEF Eastern and Southern Africa, 2023). Conflict alone displaced children from more than 14,000 schools in West and Central Africa as of 2024, and children in crisis-affected countries may acquire foundational skills up to six times more slowly than peers in stable settings (UNICEF Eastern and Southern Africa, 2023). The educational consequences extend well beyond physical access: children in emergency settings face trauma, loss of teachers and materials, and severe interruptions to instruction at the precise developmental stages when foundational competencies should be consolidated.

In Cameroon, ongoing socio-political conflict in the Northwest, Southwest, and Far North regions has left over 1.5 million school-age children requiring urgent educational assistance (UNICEF Cameroon, 2025). A situational analysis conducted under the Unlocking Data Initiative found that education data in the country is characterised by fragmentation, restricted access, and limited integration among data producers, making evidence-informed pedagogical decision-making particularly difficult (Pambe et al., 2025b). A subsequent national Evidence Gap Map identified a concentration of existing studies at the policy and system level but found almost no rigorous evidence on classroom-level teaching effectiveness in crisis-affected settings, and not a single systematic review on the topic (Pambe et al., 2025a).

Mitigating learning loss during and after emergencies demands instructional responses that are qualitatively different from those suited to stable, well-resourced contexts. Several approaches have been developed and piloted, including structured pedagogy, in which teachers are supported through scripted lesson guides and structured materials; Teaching at the Right Level (TaRL), which groups children by demonstrated competency rather than age and focuses instruction on foundational skills; and accelerated education programmes, designed for over-age and out-of-school learners (INEE, 2016; J-PAL, 2022). TaRL in particular has produced some of the largest effect sizes in education research globally, yet recent evidence from African settings

suggests its effectiveness is highly context-dependent, with implementation fidelity and the quality of teacher support emerging as critical moderators of impact (J-PAL, 2022).

A preliminary search of PROSPERO, the Cochrane Database of Systematic Reviews, the JBI Evidence Synthesis database, and the Campbell Collaboration Library did not identify a completed or underway systematic review specifically addressing the effectiveness of teaching approaches for foundational learning in emergency contexts within Sub-Saharan Africa. Existing global syntheses on education in emergencies tend to focus on access, protection, and psychosocial support, with comparatively limited synthesis of classroom-level pedagogical effectiveness (Burde et al., 2017). Where evidence syntheses on foundational learning pedagogy exist, they are predominantly drawn from stable, non-crisis contexts and may not generalise to classrooms affected by displacement, insecurity, or chronic interruption. This review addresses that gap directly.

This review responds to a Learning Agenda priority question co-developed by government officials, researchers, civil society organisations, and education practitioners through the Northwest Region Community of Practice established under the Unlocking Data Initiative in Cameroon (Fotso et al., 2025). The question was adopted precisely because stakeholders could not point to reliable, synthesised evidence on which teaching approaches work in crisis-affected classrooms. Findings are intended to inform teacher training design, classroom support guidance, and resource allocation decisions within Cameroon and across the wider set of conflict and crisis-affected contexts in Sub-Saharan Africa.

## **Review question**

What is the effectiveness of teaching approaches on foundational literacy and numeracy outcomes among primary school-age learners (5-17 years) in emergency contexts in Sub-Saharan Africa?

## **Specific sub-questions:**

1. What is the effect of different teaching approaches on foundational literacy and numeracy outcomes for learners in emergency and crisis-affected contexts in Sub-Saharan Africa, compared to standard or non-adapted teaching approaches?
2. Which teaching approaches, including structured pedagogy, teaching at the right level, accelerated and remedial instruction, psychosocial-integrated teaching, and multigrade and multilingual approaches, are most consistently associated with improved foundational learning outcomes in emergency settings?
3. What is the effect of crisis-adapted teaching approaches on learner participation, attendance, and engagement in foundational education during emergencies?
4. What contextual, implementation, or teacher-related factors explain variation in the effectiveness of teaching approaches across different emergency and crisis contexts?

## **Inclusion criteria**

Eligibility for this review is defined using the PICO framework (Participants, Interventions, Comparators, Outcomes), consistent with JBI methodology for systematic reviews of effectiveness.

### **Participants**

The review will include studies of children and adolescents enrolled or eligible for enrolment in primary or early-grade foundational education in Sub-Saharan Africa, in settings affected by armed conflict, forced displacement, natural disaster, epidemic-related disruption, or other humanitarian crisis. This includes children enrolled in formal school systems, temporary learning spaces, community-based education initiatives, and accelerated education programmes for over-age learners. The age range is broadly defined to reflect the range of learners typically addressed in emergency education programmes, from the early grades of primary schooling through to late primary or equivalent level for over-age learners in accelerated programmes. Studies focused exclusively on tertiary education, technical and vocational training, or adult literacy programmes will be excluded.

### **Interventions**

The review will include studies examining teaching approaches and pedagogical interventions adapted or applied in emergency contexts. Eligible interventions include, but are not limited to, structured pedagogy (teacher-led instruction supported by scripted lesson guides and structured learning materials), teaching at the right level (grouping learners by demonstrated competency and delivering targeted instruction at their actual learning level), accelerated and remedial learning programmes (intensive catch-up instruction for over-age, out-of-school, or learning-delayed children), multigrade instruction (teaching strategies designed for classrooms with learners at different grade levels), multilingual instruction (approaches that use children's home languages alongside official languages of instruction), distance and remote learning (including radio, television, and digital platforms used during emergencies), and psychosocial-integrated teaching (pedagogical approaches that explicitly incorporate trauma-informed or psychosocial support elements into classroom instruction). Studies examining professional development interventions for teachers as the primary mechanism for changing teaching practice will also be included where the study clearly links professional development to classroom teaching behaviour and to learner outcomes.

### **Comparators**

The review will consider studies that compare an eligible teaching approach against standard or non-adapted teaching approaches, business-as-usual instruction, no structured teaching, a waiting list or untreated comparison group, or an alternative crisis-adapted teaching approach. Both passive comparators (no intervention or usual practice) and active comparators (an alternative teaching approach) will be accepted. Studies without any comparator group will be

considered only for the qualitative complementary synthesis of implementation and contextual factors.

### **Outcomes**

The primary outcomes of interest are foundational literacy proficiency (including early grade reading, reading fluency, reading comprehension, phonological awareness, decoding, and print awareness) and foundational numeracy proficiency (including early grade mathematics, number sense, arithmetic skills, and mathematical proficiency). Secondary outcomes include learner attendance, classroom participation and engagement, school retention and dropout rates, and socio-emotional wellbeing or resilience as it relates to learning. Where studies report multiple outcomes, all eligible outcomes will be extracted and reported.

### **Exclusion Criteria**

This systematic Review will exclude studies published in languages other than English and French, Studies published before 2000, Studies conducted outside SSA, studies who's participants are below 5 years of age or above 17 years of age, observational studies and expert opinion.

### **Types of studies**

Experimental, and quasi-experimental study designs will form the primary evidence tier for the quantitative synthesis. This includes randomised controlled trials, cluster randomised controlled trials, non-randomised controlled trials, controlled before-and-after studies, interrupted time series studies, and other quasi-experimental designs with a clearly defined comparison group. Where experimental or quasi-experimental evidence is limited, high-quality observational studies with a comparison group will also be considered and clearly distinguished in the synthesis.

Qualitative studies, including studies employing interviews, focus groups, observation, or case study approaches, will be included for a complementary synthesis addressing sub-question 4 on the contextual, implementation, and teacher-related factors that explain variation in effectiveness. The qualitative and quantitative syntheses will be conducted separately and integrated at the discussion stage.

Studies published in English and French will be included. Studies published in other languages without an available English or French abstract or full text will be excluded. Studies published from the year 2000 onwards will be included. This date restriction is justified by the significant evolution of evidence-based teaching approaches for emergency education following the adoption of the INEE Minimum Standards in 2004 and the expansion of structured pedagogy and teaching at the right level approaches from the early 2000s. Studies published before 2000 are unlikely to reflect current implementation realities or contemporary understanding of effective emergency pedagogy.

Studies conducted within Sub-Saharan Africa, as defined by the World Bank regional classification, will be included. Studies conducted exclusively outside Sub-Saharan Africa will be excluded. Studies conducted in mixed or multi-country designs will be included if sufficient data are reported for Sub-Saharan African contexts to allow extraction.

## Methods

### Search strategy and information sources

The search strategy will be developed in accordance with JBI methodology for systematic reviews and will aim to identify published and unpublished evidence on the effectiveness of teaching approaches on foundational learning in emergency contexts. A comprehensive three-step approach will be followed.

In the first step, a preliminary search will be conducted in ERIC and Web of Science to identify relevant articles. Text words in titles and abstracts and index terms used to describe relevant articles will be analyzed to inform the development of the full search strategy.

In the second step, the full search strategy will be applied to eight electronic databases: Education Resources Information Center (ERIC), Open Alex, Web of Science Core Collection, Scopus, PsycINFO, Embase, Education Source, and African Journals Online (AJOL). Web of Science Core Collection, Scopus, PsycINFO, Embase, and Education Source require institutional subscription access; where this is not available to the review team, the gap will be documented transparently in the review's search log, and OpenAlex, together with a supplementary manual search of Google Scholar (first 200 relevant results, or a defensible relevance-ranked slice where the on-topic hit rate is unusually high), will serve as the review's open-access substitutes for citation-index coverage. AJOL's own search interface disallows automated, scripted querying under its robots.txt, but is directly reachable through a standard browser session; where reachable this way it will be queried directly, with the top relevance-ranked page(s) fully screened given its very large, effectively unbounded result counts for broad multi-term queries. Where only scripted access is available to support the search, a site-restricted web search (via DuckDuckGo, since Google returns a CAPTCHA challenge on repeated site-restricted queries and Bing silently drops both site: filters and quoted phrase terms) will substitute for AJOL's own search tool. These databases were selected to provide broad coverage of education, social sciences, psychology, public health, and African research, and to reduce the risk of missing relevant literature at the intersections of these fields.

In the third step, the reference lists of all included studies and relevant systematic reviews will be manually screened to identify additional eligible studies. Forward citation searching will also be conducted for key included papers.

To minimize publication bias and capture relevant unpublished evidence, grey literature sources will also be searched. These include Education Cannot Wait, the Inter-Agency Network for Education in Emergencies (INEE), UNICEF, UNESCO, the World Bank Open Knowledge Repository, the United Nations High Commissioner for Refugees (UNHCR), the Abdul Latif Jameel Poverty Action Lab (J-PAL), the Unlocking Data Initiative's own Evidence Library (unlockingdata.africa), the Association for the Development of Education in Africa (ADEA), and humanitarian agency and NGO report repositories. Should ADEA's own site search tool not be returning results at the time of searching, its content will instead be accessed by manual browsing of its publications listings. Relevant doctoral dissertations and theses will also be sought.

The search strategy is structured around the four PICO elements, translated into four Boolean search blocks combined using AND operators. The sample search strategy below is adapted for ERIC. Search strings for all other databases will be adapted to their specific indexing and thesaurus structures and will be documented and reported in the completed review. Reviewers should note that ERIC's own search interface performs loose, word-level matching within indexed description fields rather than strict phrase-level Boolean logic; a single literal Boolean string combining several multi-word phrase blocks may therefore return results that share little topical relevance with the query as written, and the search should be supplemented with a series of narrower, single-concept-pair queries rather than relying on the yield of the full literal string alone.

*("education in emergencies" OR "emergency education" OR "humanitarian education" OR "refugee education" OR "conflict-affected education" OR "education humanitarian" OR "displacement education" OR "crisis education" OR "fragile contexts education") AND ("teaching" OR "pedagogy" OR "pedagogical" OR "instruction" OR "learning" OR "classroom" OR "teacher" OR "curriculum" OR "structured pedagogy" OR "TaRL" OR "teaching at the right level" OR "accelerated learning" OR "remedial" OR "multigrade" OR "distance learning") AND ("Africa" OR "African" OR "Sub-Saharan" OR "Cameroon" OR "Kenya" OR "Nigeria" OR "Ethiopia" OR "Uganda" OR "Tanzania" OR "Ghana" OR "Mali" OR "Niger" OR "Chad" OR "Sudan" OR "Somalia" OR "Malawi" OR "Zambia" OR "Mozambique" OR "Zimbabwe" OR "Rwanda" OR "Burundi" OR "Congo")*

The search will be limited to studies published from 2000 onwards and to English and French language publications. All retrieved records will be imported into Rayyan for deduplication and screening. Full search strategies for all databases, including all search terms, Boolean operators, truncation symbols, and applied limits, will be reported in the completed review.

### **Study selection**

Following deduplication of retrieved records, a two-stage screening process will be employed by two independent reviewers.

At the first stage, two reviewers will independently screen all records at title and abstract level against the predefined PICO inclusion and exclusion criteria. Records that clearly do not meet the eligibility criteria will be excluded. Records that appear potentially eligible or where eligibility cannot be determined from the title and abstract alone will be retained for full-text review. Before commencing screening, both reviewers will conduct a calibration exercise on a sample of 25 records drawn from the database results to align their interpretation of the eligibility criteria. Discrepancies will be resolved through discussion.

At the second stage, full texts of all records retained from Stage 1 will be retrieved and assessed independently by the same two reviewers against the full set of inclusion criteria. Reasons for exclusion at full-text stage will be recorded for each excluded study and will be reported in the PRISMA 2020 flow diagram in the completed review.

Disagreements at either stage that cannot be resolved through discussion between the two primary reviewers will be referred to by an independent adjudicator. The results of the study selection process will be presented in a PRISMA 2020 flow diagram in the final review.

### **Assessment of methodological quality**

Eligible studies will be critically appraised by two independent reviewers using the standardised JBI critical appraisal instruments appropriate to each study design. For randomised controlled trials, the JBI Critical Appraisal Checklist for Randomised Controlled Trials will be applied. For quasi-experimental studies, the JBI Critical Appraisal Checklist for Quasi-Experimental Studies will be applied. For observational studies included in the quantitative synthesis, the JBI Critical Appraisal Checklist for Analytical Cross-Sectional Studies or the JBI Checklist for Cohort Studies will be applied as appropriate. For qualitative studies included in the complementary synthesis, the JBI Critical Appraisal Checklist for Qualitative Research will be applied.

Each study will be assessed item by item, and a response of yes, no, unclear, or not applicable will be assigned for each criterion. Studies will not be excluded from the review solely based on methodological quality. Rather, the results of critical appraisal will be used to contextualise the interpretation of findings in the narrative synthesis and, where applicable, in the GRADE certainty of evidence assessments. All appraisal results will be reported in narrative form and in a summary table in the completed review.

Any disagreements between reviewers during critical appraisal will be resolved through discussion. Where consensus cannot be reached, the adjudicator will be consulted. Authors of papers will be contacted to request missing or additional data where this information is required to complete the appraisal.

### **Data extraction**

Data will be extracted from included studies by two independent reviewers using the standardised data extraction form developed for this review (see Appendix A). The form will be pilot tested on three included studies before full extraction begins. Any modifications identified during piloting will be agreed by the full team before proceeding.

For each included study, the following categories of information will be extracted: source characteristics (authors, year, country, publication type); study design and methodological approach; participant characteristics (age, gender where reported, emergency context type, educational setting); intervention description (type of teaching approach, mode of delivery, intensity and duration, who delivered the intervention, and level of implementation support); comparator description; outcome measures and measurement instruments; effect estimates and associated confidence intervals for primary and secondary outcomes; disaggregated findings by gender, disability, displacement status, or other equity dimensions where reported; implementation factors (facilitators, barriers, fidelity of implementation); and key findings and conclusions.

Any disagreements between reviewers during data extraction will be resolved through discussion or referral to the adjudicator. Authors of papers will be contacted where data appear to be missing or where clarification is needed. The assumptions made about any missing or unclear data will be described in the completed review.

## **Data synthesis**

### ***Quantitative synthesis***

Studies with sufficiently comparable participants, interventions, comparators, and outcomes will be pooled in statistical meta-analysis. Effect sizes will be expressed as standardised mean differences for continuous outcomes and as odds ratios or risk ratios for dichotomous outcomes, with 95% confidence intervals reported for all estimates. Statistical heterogeneity will be assessed using the I-squared statistics and the Cochrane Q (chi-squared) test. A random-effects model will be used as the primary analytical approach given the anticipated clinical and methodological heterogeneity across emergency contexts, settings, and teaching approaches. A fixed-effect model will be considered as a sensitivity analysis where study sets are judged sufficiently homogeneous.

Where statistical pooling is not appropriate due to substantial heterogeneity or heterogeneity in outcome measurement, a narrative synthesis will be conducted. The narrative synthesis will describe the characteristics of included studies, summarise intervention components and implementation strategies, and compare findings across studies in a structured and transparent manner. Results will be presented in text and summary tables.

Subgroup analyses will be conducted, where data permit, to explore potential sources of heterogeneity. Pre-specified subgroups include type of teaching approach (structured pedagogy, TaRL, accelerated learning, multigrade, multilingual, distance learning, and psychosocial-integrated), type of emergency context (armed conflict, forced displacement, natural disaster, disease outbreak), and where data are available, learner gender and disability status. Sensitivity analyses will be conducted to assess the robustness of findings by excluding studies with a high risk of bias.

Publication bias will be assessed using funnel plot asymmetry and Egger's test where a minimum of ten studies is included in any meta-analytic pool.

### ***Qualitative synthesis***

Qualitative studies and qualitative components of mixed-methods studies will be synthesised separately using a thematic synthesis approach to address sub-question 4 on contextual, implementation, and teacher-related factors. Findings will be coded inductively and grouped into descriptive and analytical themes. The qualitative and quantitative syntheses will be integrated at the discussion stage to provide a more complete account of both what works and why it works in specific emergency education contexts.

### ***Certainty of evidence***

The certainty of the evidence for each primary and secondary outcome will be assessed using the Grading of Recommendations Assessment, Development and Evaluation (GRADE) approach (Guyatt et al., 2008). Two reviewers will independently assess each outcome, with disagreements resolved through discussion. The certainty of evidence will be rated as high, moderate, low, or very low based on assessments of risk of bias, inconsistency, indirectness, imprecision, and publication bias for quantitative outcomes.

A Summary of Findings table will be produced using GRADEpro GDT software. The table will present the key effect estimates, the number of participants and studies contributing to each estimate, and the overall certainty of evidence for each major outcome. The table will provide a concise, transparent summary of the strength of the evidence base for the review's principal conclusions.

### **Projected timeline**

The review is structured within the Northwest Region CoP DBIR Cycle 3 implementation plan, which runs from May to November 2026.

- May to June 2026: Protocol finalisation and PROSPERO registration, prior to commencing any database searches.
- June to July 2026: Database searches across all eight academic databases and grey literature sources.
- July 2026: Title and abstract screening, following completion of calibration exercise.
- August 2026: Full-text screening, with reasons for exclusion documented for each excluded study.
- August to September 2026: Critical appraisal of all eligible studies and data extraction using the standardised form.
- September to October 2026: Meta-analytic and/or narrative synthesis, GRADE assessments, and Summary of Findings table.
- October 2026: Draft manuscript preparation and peer feedback at Workshop 3.
- October to November 2026: Final manuscript revision, submission, and presentation at the CoP plenary.

### **Notable deviations from the methodological approach**

This protocol sets out the planned methodology to ensure transparency and reproducibility. All methods described will be adhered to in accordance with JBI guidance. However, systematic reviews are inherently iterative, and adjustments may become necessary in response to the volume and characteristics of the retrieved literature.

Any modifications to the methods described in this protocol will be carefully documented, clearly justified, and reported transparently in the final review manuscript under a dedicated Deviations from Protocol subsection. Potential deviations may include broadening or narrowing the scope of eligible study designs if the initial searches reveal a substantially larger or smaller evidence base than anticipated, modifying the planned subgroup analyses if the data are insufficient to support pre-specified comparisons, or adjusting the GRADE assessments in response to the specific study designs and outcome reporting found in the included evidence. No deviations are anticipated at the protocol stage.

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## **Appendix A: Draft Data Extraction Form**